

TECHBIG Solutions

SAP S/4HANA Materials Management (MM) Complete E2E Implementation Program

Course Positioning: From Procurement Fundamentals to S/4HANA Real-Time MM Implementation

Recommended Methodology: 30% Theory + 70% Hands-on

Learning Cycle: Learn → Configure → Execute → Integrate → Test → Troubleshoot → Implement

COMPLETE SAP S/4HANA MM SYLLABUS

MODULE 1 – SAP S/4HANA & SAP MM INTRODUCTION

- SAP ERP overview; SAP ECC vs S/4HANA; SAP S/4HANA architecture; HANA database; Fiori; MM within S/4HANA Enterprise Management.
- Procurement and inventory business processes; consultant vs business-user roles; end-to-end Procure-to-Pay overview.
- Key S/4HANA changes relevant to MM: Business Partner, Material/Product Master, simplified data model, Fiori and analytics.

MODULE 2 – ORGANIZATIONAL STRUCTURE

- Define Company and Company Code; Plant; Storage Location; Purchasing Organization; Purchasing Group.
- Assignments between company code, plant, purchasing organization and purchasing group.
- Centralized vs decentralized purchasing; reference purchasing organizations; enterprise structure design.

MODULE 3 – MATERIAL / PRODUCT MASTER

- Material/Product Master concept; material types; industry sector; number ranges; material groups.
- Views: Basic Data, Purchasing, MRP, Accounting, Storage, Quality and Sales-related views.
- Units of measure; valuation class; price control; batch management; serial numbers; deletion/block indicators.
- Material master creation, extension, change and display; mass maintenance and common data issues.

MODULE 4 – SUPPLIER / BUSINESS PARTNER MASTER

- Business Partner concept in S/4HANA; supplier roles; account groups; number ranges.
- General data, company-code data and purchasing data; reconciliation account; payment terms; partner functions.
- Supplier creation/change/display and master-data governance.

MODULE 5 – PROCUREMENT MASTER DATA

- Purchasing Info Records; source lists; quota arrangements; conditions; source determination.
- Supplier-material relationships; purchasing organization data; pricing conditions and validity.
- Source determination and automatic source assignment.

MODULE 6 – MATERIAL VALUATION & ACCOUNT ASSIGNMENT

- Valuation area; valuation class; price control; standard price vs moving average price.
- Automatic account determination; valuation grouping; transaction keys; OBYC concept.
- Account assignment categories: cost center, asset, order, project and sales-order related scenarios.
- Material valuation, split valuation and accounting integration.

MODULE 7 – PURCHASE REQUISITION (PR)

- PR concept and lifecycle; creation manually and from MRP; item categories; account assignment.
- PR release strategy/workflow; source assignment; conversion to RFQ/PO; monitoring and reporting.
- Fiori procurement apps and requisitioning concepts.

MODULE 8 – REQUEST FOR QUOTATION & QUOTATION

- RFQ creation; vendor selection; quotation maintenance; price comparison; quotation evaluation.

- Source determination, quotation comparison and conversion to purchase order.

MODULE 9 – PURCHASE ORDER (PO)

- PO types; document types; number ranges; item categories; account assignment categories.
- Standard PO; stock material; consumable procurement; service PO; subcontracting; consignment; third-party scenarios.
- PO pricing conditions; texts; confirmations; delivery tolerances; invoice tolerances; output/message determination.
- PO release strategies/workflows and approval processes.

MODULE 10 – PURCHASE CONTRACTS & SCHEDULING AGREEMENTS

- Quantity/value contracts; contract release orders; validity and target quantities/values.
- Scheduling agreements; schedule lines; release documentation; monitoring.
- Conditions, source determination and practical procurement scenarios.

MODULE 11 – PURCHASING CONFIGURATION

- Purchasing document types; screen/layout control; field selection; number ranges; item categories.
- Account assignment; allowed combinations; tolerance settings; purchasing groups and organizations.
- Release procedures/workflows; output management; text determination; confirmation control.

MODULE 12 – INVENTORY MANAGEMENT

- Goods movements and movement types; stock types; stock overview.
- Goods receipt against PO; GR without PO; stock transfer; transfer posting; goods issue; reversals.
- Unrestricted, quality inspection and blocked stock; reservations; stock adjustments.
- Physical inventory: documents, counting, differences, recounting and posting.

MODULE 13 – SPECIAL STOCK & SPECIAL PROCUREMENT

- Consignment stock; subcontracting; pipeline material; project stock overview.
- Subcontracting process: components, subcontracting PO, component issue, GR and consumption.
- Vendor consignment: receipt, consumption, settlement and transfer to own stock.

MODULE 14 – INVOICE VERIFICATION

- Logistics Invoice Verification; invoice entry with PO reference; planned/unplanned delivery costs.
- Three-way match: PO, GR and invoice; quantity/price variances; blocked invoices.
- Credit memo, subsequent debit/credit, invoice cancellation and reversal.
- Automatic invoice settlement concepts and GR-based invoice verification.

MODULE 15 – AUTOMATIC ACCOUNT DETERMINATION

- Valuation class, account category reference and transaction keys.
- OBYC configuration concepts: inventory, GR/IR, consumption, price differences and freight.
- Debugging account determination errors and FI-MM integration.

MODULE 16 – STOCK TRANSFER & STOCK TRANSPORT

- Plant-to-plant transfers; storage-location transfers; one-step and two-step procedures.
- Stock Transport Orders; intra-company and inter-company concepts; delivery-based STO overview.
- Goods issue, goods receipt, stock in transit and accounting impact.

MODULE 17 – MRP & MM INTEGRATION

- MRP overview; material master MRP views; planning parameters; procurement proposals.
- Planned orders, purchase requisitions and external procurement.
- MRP-generated procurement, source determination and conversion into purchasing documents.

MODULE 18 – BATCH MANAGEMENT & SERIAL NUMBERS

- Batch master; batch creation; batch classification; batch determination; shelf-life management.
- Serial number profiles; serial number assignment; goods movements with serialized materials.
- Practical batch/serial scenarios and troubleshooting.

MODULE 19 – PHYSICAL INVENTORY

- Physical inventory process; inventory documents; counting; recount; differences.
- Posting inventory differences; cycle counting overview; inventory reports and controls.

MODULE 20 – SERVICES PROCUREMENT

- Service master/product concept; service specifications; service purchase orders.
- Service entry sheets; acceptance; invoice verification; account assignment.
- Planned and unplanned services; service procurement scenarios.

MODULE 21 – SOURCE DETERMINATION & SOURCING

- Source lists; quota arrangements; purchasing info records; source determination.
- Automatic source assignment; fixed sources; preferred suppliers.
- Supplier evaluation and sourcing optimization.

MODULE 22 – WORKFLOWS & APPROVALS

- Flexible workflows in procurement; approval rules; release workflows.
- PR and PO approval scenarios; workflow troubleshooting; substitutes and escalation concepts.

MODULE 23 – PROCUREMENT ANALYTICS & REPORTING

- Purchase requisition and PO monitoring; supplier spend; purchasing analytics.
- Material documents; stock reports; inventory valuation; invoice verification reports.
- Supplier evaluation; procurement overview; Fiori analytical apps and KPIs.

MODULE 24 – FI-MM INTEGRATION

- Procure-to-Pay accounting flow: PO → GR → IR → Payment.
- GR accounting: Inventory Dr / GR-IR Cr; IR accounting: GR-IR Dr / Vendor Cr.
- Account determination, valuation class, tax, GR/IR clearing and invoice variances.
- Integration troubleshooting between MM and FI.

MODULE 25 – MM-SD & CROSS-MODULE INTEGRATION

- Material master integration with Sales; stock availability; goods issue and inventory impact.
- MM-PP integration: BOM, production consumption and procurement.
- MM-QM overview: inspection stock and quality-related procurement.

- MM-WM/EWM overview: warehouse movements and stock integration.

MODULE 26 – TAX, GST & INDIA LOCALIZATION OVERVIEW

- GST concepts relevant to procurement; input tax; CGST/SGST/IGST.
- Tax codes and condition-based tax determination overview; reverse charge scenarios.
- Invoice verification and tax validation; India procurement compliance overview.

MODULE 27 – FIORI FOR PROCUREMENT

- Fiori Launchpad; spaces/pages; business roles and catalogs.
- Manage Purchase Requisitions; Manage Purchase Orders; Monitor Purchase Order Items; supplier/material apps.
- Stock overview; material documents; procurement analytics and approval applications.

MODULE 28 – S/4HANA PROCUREMENT FEATURES

- Business Partner and Product Master; simplified procurement processes; Fiori-first execution.
- Flexible workflows; situation handling; automated procurement; supplier evaluation and spend visibility.
- Procurement APIs and integration concepts; public vs private edition overview.

MODULE 29 – SAP IMPLEMENTATION METHODOLOGY

- SAP Activate: Prepare, Explore, Realize, Deploy and Run.
- Fit-to-standard / requirement gathering; As-Is and To-Be; gap analysis; configuration; testing; migration; cutover; go-live and hypercare.
- MM project documentation: BRD, BPD, configuration document, functional specification, test scripts, defect log, cutover checklist and support matrix.

MODULE 30 – DATA MIGRATION

- Material master migration; supplier/BP migration; purchasing info records; opening stock.
- Migration templates and data validation; legacy-to-S/4HANA mapping; reconciliation and mock loads.
- Migration cockpit concepts and migration testing.

MODULE 31 – TESTING

- Unit testing; System Integration Testing; User Acceptance Testing; regression testing.
- End-to-end P2P test scenarios; negative testing; defect lifecycle; retesting and sign-off.

MODULE 32 – PRODUCTION SUPPORT

- Incident priorities, SLA, root-cause analysis and functional troubleshooting.
- Common MM issues: PO errors, release failures, source determination, GR/IR, account determination, invoice blocks, stock discrepancies and valuation errors.
- Support documentation, ticket analysis and business-user communication.

MODULE 33 – REAL-TIME E2E BUSINESS SCENARIOS

- Standard Procure-to-Pay: PR → PO → GR → Invoice → Payment.
- Consumable procurement; stock procurement; service procurement.
- Subcontracting; consignment; STO; physical inventory; returns to vendor.
- Complete FI-MM integration and month-end procurement reconciliation.

MODULE 34 – FINAL CAPSTONE PROJECT

- Implement SAP S/4HANA MM for TECHBIG Global Solutions Pvt. Ltd.
- Configure enterprise structure, material/product master, Business Partner/supplier, purchasing, inventory, valuation, account determination, invoice verification, workflows and reporting.
- Execute complete P2P, subcontracting, consignment, STO, physical inventory and FI-MM integration scenarios.
- Deliverables: BRD, BPD, configuration document, master-data templates, test scripts, defect log, cutover plan, go-live checklist and support matrix.

MODULE 35 – INTERVIEW & PRACTICAL ASSESSMENT

- MM fundamentals; enterprise structure; master data; purchasing; inventory; valuation; account determination; invoice verification; special procurement; MRP integration; FI integration; S/4HANA changes.
- Hands-on assessments: material creation, BP/supplier, PR, RFQ, PO, GR, invoice, stock transfer, physical inventory and troubleshooting.
- Final assessment: complete E2E implementation case study.

RECOMMENDED COURSE DURATION

Option A – Job-Oriented: 16 Weeks

Weeks 1–2: SAP/S4HANA + MM fundamentals | Weeks 3–5: Master Data & Enterprise Structure | Weeks 6–9: Purchasing | Weeks 10–12: Inventory & Invoice Verification | Week 13: Integration | Week 14: Advanced Procurement | Week 15: Implementation/Testing | Week 16: Capstone + Interview.

Option B – Intensive: 12 Weeks

5–6 days per week with strong hands-on practice.

Option C – Professional E2E: 20 Weeks

Includes configuration, integration, migration, testing, implementation documentation, go-live and production support.

STUDENT OUTCOMES

After completing the program, learners should be able to configure and execute SAP S/4HANA MM processes, manage material and supplier master data, execute purchasing and inventory transactions, configure valuation and account determination, perform invoice verification, work with Fiori procurement applications, understand FI/MM and cross-module integration, participate in implementation, testing, migration, go-live and production support.

TECHBIG SOLUTIONS – COURSE USP

Course Name: SAP S/4HANA MM – E2E Implementation & Real-Time Project Training

Target Audience: SAP MM beginners, procurement professionals, supply-chain professionals, commerce graduates, SAP ECC MM professionals upgrading to S/4HANA and IT professionals moving into SAP MM.

Final Certificate: TECHBIG Solutions – Certificate in SAP S/4HANA Materials Management (MM) E2E Implementation